Python Programming Assignment

Objective

The purpose of this assignment is to develop a strong understanding of basic Python programming concepts through practical implementation. The exercises focus on problem-solving using control structures, loops, functions, and data structures such as lists and dictionaries.

Exercise 1: Prime Number Check

Problem Statement

To write a Python program that determines whether a given number is a prime number.

Approach

A number is considered prime if it is greater than 1 and divisible only by 1 and itself. The program checks divisibility from 2 up to the square root of the number to improve efficiency.

Result

The program correctly identifies whether a number is prime or not.
Example: 29 → Prime Number

Exercise 2: Product of Random Numbers

Problem Statement

To generate two random numbers and verify if the user correctly calculates their product.

Approach

Two random integers are generated using the random module. The user is prompted to input the product, and the program checks whether the answer matches the actual result.

Result

The program successfully validates user input and displays whether the answer is correct or incorrect.

Exercise 3: Squares of Even Numbers (100–200)

Problem Statement

To print the squares of either even or odd numbers within the range of 100 to 200.

Chosen Option: Even Numbers

Approach

A loop is used to iterate through the range. A condition checks whether the number is even, and its square is calculated and displayed.

Result

All even numbers between 100 and 200 are squared and printed successfully.

Exercise 4: Word Counter

Problem Statement

To count the frequency of each word in a given text.

Approach

The input text is split into individual words. A dictionary is used to store each word along with its frequency count by iterating through the list of words.

Result

The program correctly counts and displays the number of occurrences of each word in the text.

Exercise 5: Palindrome Check

Problem Statement

To check whether a given string is a palindrome.

Approach

The string is converted to lowercase and cleaned by removing spaces and non-alphanumeric characters. The processed string is then compared with its reverse.

Result

The program accurately determines whether a string is a palindrome.
Example: “racecar” → True

Conclusion

This assignment helped in understanding and applying key Python programming concepts such as:

- Conditional statements
- Looping constructs
- Functions

- String manipulation
- Dictionary usage
- Basic problem-solving techniques

The exercises improved logical thinking and demonstrated how Python can be used to solve real-world computational problems efficiently.